1. **Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.**

- To divide the network 192.168.10.0/24 into 4 equal subnets:
- Original CIDR is : /24 which means its subnet mask is 255.255.255.0
- To create 4 subnets, I need to borrow 2 bits from the 8 host bits.
- Now, New CIDR is : /26 which means its subnet mask it 255.255.255.192
- Each /26 CIDR has: 64 IP addresses and 62 usable host addresses.

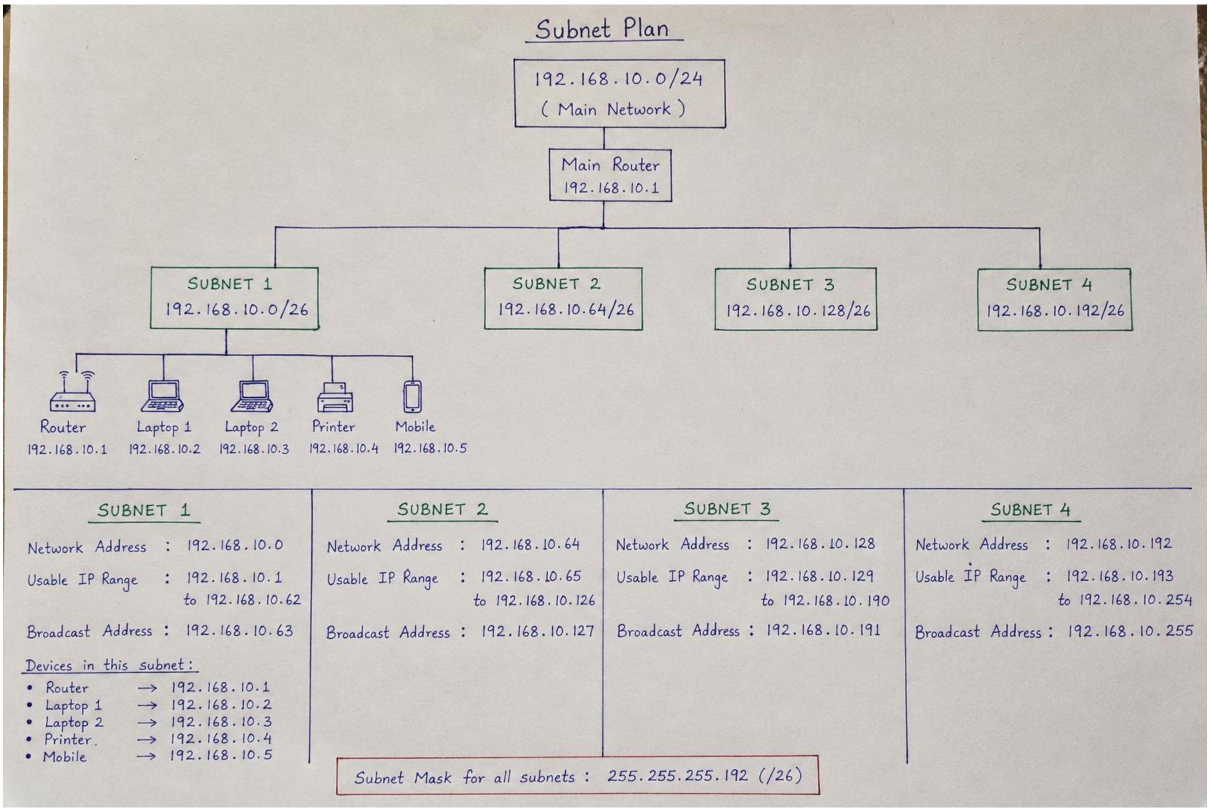
Subnet	Subnet Mask	Network Address	Broadcast Address
1	255.255.255.192/26	192.168.10.0	192.168.10.63
2	255.255.255.192/26	192.168.10.64	192.168.10.127
3	255.255.255.192/26	192.168.10.128	192.168.10.191
4	255.255.255.192/26	192.168.10.192	192.168.10.255

2. **For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.**

- Using one of the subnets created from 192.168.10.0/24 dividing into 4 equal subnets. Here, I choose first subnet:
- Subnet is 192.168.10.0/26
- Subnet mask is 255.255.255.192
- Network address is 192.168.10.0
- Broadcast address is 192.168.10.255
- And Valid host range is 192.168.10.1 – 192.168.10.62

Device	**Assigned IP address**
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5

3. **Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.**



4. **Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.**

- Here, I am explaining subnetting to a friend who has never done it before.

Step 1 Starting with the Original Network

- I was given Network address – 192.168.10.0/24 and subnet mask is 255.255.255.0.
- /24 means the first 24 bits are used for the network and 8 bits for the host devices.

Step 2 Divide the Network into 4 Equal Subnets

- I needed 4 equal subnets.
- To create 4 subnets, I borrowed 2 bits from the host bits because: 2^1 = 2 which is not enough, 2^2=4 subnets which are enough.
- So the subnet mask changes from /24 to /26 now and new subnet mask is now 255.255.255.192

Step 3 Calculation of Number of Hosts per Subnet

- After borrowing 2 bits, there are: 32 -26 = 6 host bits.
- Now I am calculating usable hosts 2^6 – 2 = 64 and 64-2-= 62 usable hosts. From here, I removed 2 addresses which are reserved for the network address and broadcast address.
- So each subnet has 64 total IP addresses and 62 useable host IP addresses.

Step 4 finding the subnet ranges

- /26 subnet increases by 64 in the last octet.
- So the four subnets are :

Subnet	**Network Address**	**Usable Host Range**	**Broadcast Address**
1	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 - 192.168.10.126	192.168.10.127
3	192.168.19.128	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 - 192.168.10.254	192.168.10.255

Step 5 Assigning IP addresses to Devices

- For the first subnet 192.168.10.0/26 I assigned IP address to five devices.

Device	**Assigned IP Address**
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5